

Name _____

Period _____ Date _____

Practice Test C – Advanced Kinematics

On any problems, you must show all work, include units, round properly, and box your final answer.

Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

(25%) A ball is thrown horizontally at 6.0 m/s from the top of a building that is 19.6 m tall. Air resistance is negligible. Up is the positive direction. On separate axes, sketch (a) the ball's horizontal velocity v_x and (b) its vertical velocity v_y , each as a function of time, from the moment it is thrown until it hits the ground. Include numbers and units on both axes.

(25%)

- c. Is the shirt rising or falling when the fan catches it? Justify your answer with a calculation.

A driver traveling at 28.0 m/s slams on the brakes. The car slows with constant acceleration and stops after sliding 56.0 m.

(25%)

a. What is the car's acceleration?

b. How long does it take the car to stop?

c. How fast is the car moving after it has slid the first 28.0 m?

(25%)

- a. Derive an expression for the ball's maximum height H in terms of v_0 , θ_0 , and physical constants, as appropriate.
- b. A second ball is thrown straight up and reaches the same maximum height H . Derive an expression for its launch speed v_{up} in terms of v_0 and θ_0 .

Answer Key – Practice Test C

Fully worked. Algebra first, numbers last, units inside every parenthesis.

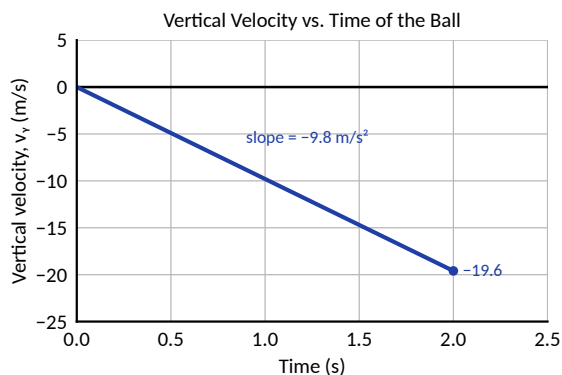
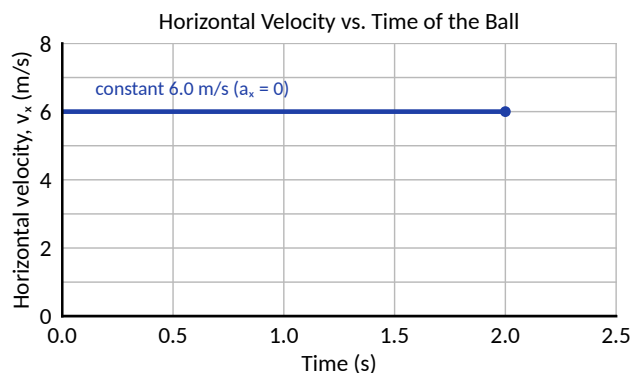
1. Sketch v_x -t and v_y -t (25%)

First find how long it is in the air (vertical, $v_{y0} = 0$, $\Delta y = -h$):

$$-h = -\frac{1}{2}gt^2 \rightarrow t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2(19.6 \text{ m})}{(9.8 \text{ m/s}^2)}} = 2.00 \text{ s}$$

(a) $a_x = 0$, so $v_x = 6.0 \text{ m/s}$ the whole time: flat line from 0 to 2.00 s

(b) $v_y = v_{y0} - gt$: starts at 0, straight line down with slope -9.8 m/s^2 , reaches $-(9.8 \text{ m/s}^2)(2.00 \text{ s}) = -19.6 \text{ m/s}$ at 2.00 s



2. T-shirt cannon (25%)

horizontal	vertical
$a_x = 0$	$a_y = -g$
$v_{x0} = v_0 \cos\theta_0$	$v_{y0} = v_0 \sin\theta_0$
$\Delta x = D = 40.0 \text{ m}$	$\Delta y = h = ?$

(a) Horizontal gives the time: $D = (v_0 \cos\theta_0)t \rightarrow t = \frac{D}{v_0 \cos\theta_0}$

$$t = \frac{(40.0 \text{ m})}{(25.0 \text{ m/s})(\cos 50.0^\circ)} = \boxed{2.49 \text{ s}}$$

(b) Same t in the vertical. Substitute the expression for t:

$$h = (v_0 \sin \theta_0)t - \frac{1}{2}gt^2 = (v_0 \sin \theta_0) \frac{D}{v_0 \cos \theta_0} - \frac{1}{2}g \frac{D^2}{v_0^2 \cos^2 \theta_0}$$

$$h = D \tan \theta_0 - \frac{gD^2}{2v_0^2 \cos^2 \theta_0}$$

$$h = (40.0 \text{ m})(\tan 50.0^\circ) - \frac{(9.8 \text{ m/s}^2)(40.0 \text{ m})^2}{2(25.0 \text{ m/s})^2(\cos 50.0^\circ)^2} = 47.7 \text{ m} - 30.4 \text{ m} = \boxed{17.3 \text{ m}}$$

$$(c) v_y = v_0 \sin \theta_0 - gt = v_0 \sin \theta_0 - \frac{gD}{v_0 \cos \theta_0}$$

$$v_y = (25.0 \text{ m/s})(\sin 50.0^\circ) - \frac{(9.8 \text{ m/s}^2)(40.0 \text{ m})}{(25.0 \text{ m/s})(\cos 50.0^\circ)} = 19.15 \text{ m/s} - 24.39 \text{ m/s} = -5.24 \text{ m/s}$$

v_y is negative, so the shirt is **falling** when it is caught (past the top of its path).

3. Braking car (25%)

Direction of motion is +. $v_0 = 28.0 \text{ m/s}$, $v = 0$, $\Delta x = d = 56.0 \text{ m}$.

$$(a) \text{ No time given or asked: } v^2 = v_0^2 + 2a\Delta x \rightarrow 0 = v_0^2 + 2ad \rightarrow a = -\frac{v_0^2}{2d}$$

$$a = -\frac{(28.0 \text{ m/s})^2}{2(56.0 \text{ m})} = \boxed{-7.00 \text{ m/s}^2} \text{ (negative: opposite the motion, slowing down)}$$

$$(b) v = v_0 + at \rightarrow t = \frac{v - v_0}{a} = \frac{(0 \text{ m/s} - 28.0 \text{ m/s})}{(-7.00 \text{ m/s}^2)} = \boxed{4.00 \text{ s}}$$

$$(c) v^2 = v_0^2 + 2a \frac{d}{2} = v_0^2 + 2 \frac{-v_0^2}{2d} \frac{d}{2} = \frac{v_0^2}{2} \rightarrow v = \frac{v_0}{\sqrt{2}}$$

$$v = \frac{(28.0 \text{ m/s})}{\sqrt{2}} = \boxed{19.8 \text{ m/s}}$$

Watch out: halfway through the distance is not half the speed (14.0 m/s). The car covers the first half fast and the second half slow.

4. Maximum height derivation (25%), variables only

(a) At the top $v_y = 0$. Time is not needed: $v_y^2 = v_{y0}^2 + 2a_y\Delta y$

$$0 = (v_0 \sin \theta_0)^2 - 2gH \rightarrow \boxed{H = \frac{v_0^2 \sin^2 \theta_0}{2g}}$$

(b) Straight up: same top condition, $v = 0$ at height H: $0 = v_{up}^2 - 2gH \rightarrow v_{up} = \sqrt{2gH}$

$$\text{Substitute H: } v_{up} = \sqrt{2g \frac{v_0^2 \sin^2 \theta_0}{2g}} = \sqrt{v_0^2 \sin^2 \theta_0} \rightarrow \boxed{v_{up} = v_0 \sin \theta_0}$$

This says the height only depends on the vertical part of the launch velocity. The horizontal motion never affects how high it goes.